Guillaume Keifer U. Arenas

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EDUCATION

De La Salle University *2021 – expected 2027*

Bachelor of Science and Master of Science in Manufacturing Engineering and Management

CGPA: 3.58/4.00; Candidate for Cum Laude; DOST-Merit and ERDT Scholar

EXPERIENCE

Graduate Research Assistant, Taiwan Experience Education Program *Aug 2025 – Dec 2025; May 2026 – Jul 2026*

National Chin-Yi University of Technology, Taichung, Taiwan GitHub

- Built a mechanoluminescent elastomer grid prototype and a Python feature-extraction pipeline converting video trials into light-intensity features for modeling; co-authored "Preliminary Evaluation of Mechanoluminescent Lattice-Patterned Sensing Skin for Visuotactile Contact Differentiation," published in IEEE ICCT 2026.
- Delivered a MediaPipe-based knee-angle baseline (hip–knee–ankle landmarks) to generate joint-angle time series for an NSTC research project.
- Engineering a camera-calibration and marker-tracking pipeline (Python, OpenCV) for a wrist-worn visuotactile grip-force sensor (GripVT); thesis targeting Sep 2026 completion.

PROJECTS

Low-Cost Automated Visuotactile Sensor Testing Rig *May – Jun 2026*

Python, OpenCV, Ender 3 (G-code) — submitted to VTCA 2026 GitHub

- Engineered a real-time computer-vision pipeline (Python, OpenCV, SciPy) tracking 150+ markers/frame at ~30 FPS via Kalman filtering and Hungarian assignment, generating 1.6M+ structured measurements per run on a self-built rig (Ender 3 3D printer retrofitted with a load cell and camera, ~98% cost reduction vs. $10K–$14K commercial systems).
- Compared 4 elastomer blends via one-way ANOVA + Tukey HSD ($p<0.001$), identifying a statistically dominant formulation and delivering a data-driven material recommendation.

Metaheuristic Feature Selection for Wearable Activity Classification *Jul 2025 – Aug 2025*

Python GitHub

- Built a benchmarking pipeline (EDA + feature engineering + baselines) across KNN, SVM, RF, XGBoost, LightGBM, and MLP on a Kaggle wearable dataset.
- Implemented and compared 8 binarized wrapper feature-selection methods (bGA, bPSO, bGWO, bHHO, bWOA, bMPA, bEO, bJAYA) to optimize accuracy with fewer inputs.
- Cut required inputs by ~85% while improving baseline model performance using the best-performing bGA feature subset.

News Summarizer Bot *Apr 2026 – Present*

Python, LLM (BERT), Web Scraping, GitHub Actions GitHub

- Built an automated pipeline that scrapes news from 3 sources, summarizes articles with an LLM (BERT-based), and delivers digests via Telegram, scheduled through GitHub Actions cron jobs.
- Sustained 97.7% run uptime and a 90.8% article-summarization success rate across 88 automated runs and 1,500+ articles over 3 months of continuous operation.

Microplastics Classification *Jan 2025 – Apr 2025*

Python

- Curated a labeled microscope image dataset (Nylon, PET, EPS) and delivered a benchmark comparing traditional ML vs. deep learning approaches; findings accepted at IEEE TENCON 2025.

Particulate Matter Prediction — Research Team Leader *Jan 2024 – Dec 2024*

De La Salle University

- Led a small research team coordinating data acquisition across 17 NCR cities and multiple organizations into one unified PM dataset.
- Built an end-to-end Python pipeline (cleaning, EDA) and trained/benchmarked MLP, Random Forest, and LightGBM baselines, documenting performance limits and next steps.

PCB Defect Detection Research *Oct – Dec 2023*

IEEE TENCON 2024, DOI: 10.1109/TENCON61640.2024.10902913

- Optimized a YOLOv8s PCB defect detector with TensorRT, achieving 98.5% mAP50 while reducing inference time ~0.9ms and model size ~39.3% versus baseline; published in IEEE TENCON 2024.

SKILLS

Programming & Data Analysis: Python, SQL, MATLAB, Pandas, NumPy, EDA, statistical analysis (ANOVA, hypothesis testing)

Machine Learning & CV: scikit-learn, TensorFlow/Keras, XGBoost/LightGBM, YOLO, OpenCV, MediaPipe, LLM integration (BERT)

AI Tools: Claude Code, Prompt Engineering, LLM API Integration

Tools & Automation: Git/GitHub, GitHub Actions, Jupyter Notebook, Matplotlib/Seaborn